

NOTES ON Gravity and Cosmology

version 1.0

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For those who come after

PREFACE

In a sense, you might say I'm getting quite fond of writing these kind of notes.

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor Omar Zanusso for the DG, Hamiltonian, and BHs parts, and on the notes of the lectures by Professors Giovanni Marozzi and Daniele Gaggero for the Cosmology part.

These notes refer to the lectures of the academic year 2025-2026, so future versions of these lectures might differ slightly.

At any rate, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer¹.

You can find all the notes I've redacted [here](#).

Without further ado, we'd better not lose much more time on a preface and get started with it.

There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur [...] But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly. Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.

*Ainulindalë, "The music of the Ainur",
Silmarillion, J. R. R. Tolkien*

¹ I'd like, however, not to see my house stormed by homing pigeons.

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1

BASIS OF DIFFERENTIAL GEOMETRY

Coming soon [1].

2 | HAMILTONIAN APPROACH

Coming soon.

3 | BLACK HOLES

Coming soon.

4

COSMOLOGY

PRE-RELATIVISTIC COSMOLOGY AND ITS PROBLEMS

Long before GR as we know it today was a thing, Cosmology was already an open field of study. Of course, in this first age of Cosmology, studies were based on Newtonian mechanics and the Newtonian theory of gravity.

Newtonian Cosmology assumed an infinite, homogeneous, and static Universe, which was also eternal and responded to Euclidean geometry. Not surprisingly, under this set of un-reasonable hypothesis¹, problems began to sprout overnight.

Consider for example the so called **Olber's paradox** (1826). Under the assumptions of Newtonian Cosmology, Olber's paradox would implicate an always-bright sky, while we have empirical evidence it isn't. In fact, the total luminosity of a set of stars with solar luminosity $L_{\odot}$ within a sphere of radius R

$$\phi_s = \int_0^R \frac{L_{\odot}}{4\pi r^2} n_s 4\pi r^2 dr$$

will tend to infinity once $R \rightarrow \infty$. That is because, under the hypothesis of homogeneity, the surface density of stars n_s cannot depend on the spatial coordinates.

Olber will then try to solve his own paradox, claiming it to be solved by dust in the ISM (read: Interstellar Medium). But, if you think about it, it really doesn't solve anything: Sooner or later, the dust itself will get into thermodynamical equilibrium with the radiation, and at that point dust will begin to emit on its own.

A second solution was given in 1872 from Zolbnev, who assumed that the Universe is *closed* with a *positive curvature*. The Universe would then be finite, and the assumption of homogeneity wouldn't let the integral above diverge.

However, solved a problem, there's always another one² peeking around the corner. In this particular case it's the problem of gravitational stability. In fact, it's evident that local instabilities in the gravitational potential must have occurred at some point in time through fluctuations of density, leading to collapses and star formation epochs. As I said, the problem is mainly local, and instabilities are generally balanced on the larger scales.

Bentley then showed that perturbations on spatial scale r have, under the assumption of homogeneity, an associated potential $\propto r^2$ and an associated acceleration $\propto r$, which leads to unstable equilibrium.

This problem is however partially mitigated by Seelinger (1895), who assumes gravity to be *shielded* on the large scales

$$\phi = G \int \frac{\rho}{r} d^3x \rightarrow \phi = G \int \frac{\rho}{r} e^{-\lambda r} d^3x$$

¹ The only reasonable assumption was for the Universe to be homogeneous and, only at times, static. The rest of the hypothesis were arguable at best.

² "Always two there are."

where the exponential term is an absorption/shielding factor. This correction will thus bring about a modification of Poisson's equation

$$\nabla^2\phi = 4\pi G\rho \rightarrow \nabla^2\phi - \lambda\phi = 4\pi G\rho$$

Much expectedly, quite a lot changed since 1917, when relativistic cosmology was officially born [2]. Einstein's GR was based on three main principles:

1. (Weak) Equivalence Principle: Inertial acceleration and gravitational acceleration are proportional to each other. Hence all free-falling systems are locally equivalent to an inertial system;
2. General Covariance Principle: When written in covariant form, the laws of Physics are the same in all reference frame;
3. Mach's Principle³ The momentum of all bodies is determined by all the masses in the Universe. Alternatively, the only things with physical sense are those that are measurable.

Only a year prior to Einstein's 1917 article, Schwarzschild had already found a solution to Einstein's celebrated equation under the assumption of a single spherical body in an otherwise empty space. Such solution, that is nowadays known as Schwarzschild's metric, has to the perk to come down to the standard spacetime Minkowski's metric at infinite distance from the object. This, of course, felt like a big problem for Mach's philosophical principle.

Einstein, perhaps conditioned from the beliefs of his time, twisted his theory so to produce a closed, static Universe. To do so, he only had to add a single term to his equation, the Cosmological Constant

$$G_{\mu\nu} - \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (1)$$

which is actually perfectly valid, even from a theoretical point of view⁴. That said, the interpretation of the Λ term was somewhat...elusive.

In 1918, Erwin Schrödinger suggested that the Cosmological Constant was equivalent to the energy-momentum tensor of a fluid with negative pressure, and could thus be included in the right-hand side term of the equation.

But, as you perhaps know, even Einstein was prone to errors like any other human being. In 1917, Vesto Slipher observed a discrepancy in the observed emission lines of some galaxies (with respect to the expected frequencies). He, however, failed to correlate such discrepancy to the distance of the aforementioned galaxies.

It was only when Hubble measured their distance [3] through the Cepheid-variables method that cosmologists started to realize that the Universe was expanding rather than being static; the expansion law of the Universe still carries Hubble's name, who was the first to suggest such correlation.

Another viable model to describe the Universe is the so-called "Einstein-De Sitter" Universe. This model essentially assumes the Universe to be akin to a non-relativistic fluid for the sake of computing the resulting metric and

³ This one is a bit "philosophical", but it's interesting to at least mention it.

⁴ I assume the four readers of these notes to be at least familiar with the terms and symbols used. If not, I suggest reading [1] or any graduate/postgraduate textbook about GR.

the properties stemming from it. This model was however abandoned because it produced errors in the estimation of the age of the Universe⁵.

Thermodynamical Considerations on the Big Bang

The Laws of Thermodynamics speak rather clearly: Energy is (globally) conserved, and Entropy either remains constant or grows. But, if entropy is not growing, the system (i.e. the Universe) doesn't evolve and will meet its end in a rather ominous-sounding way: Thermal Death.

If, on the other hand, entropy is allowed to grow, then it is possible for the system to switch to more probable microscopical configurations. So, starting from a low-entropy, extremely ordered state, the Universe would then evolve through thermodynamical means into the messy Universe we see today.

4.1 MAXIMALLY SYMMETRIC UNIVERSES

Note: Parts of this section will be eventually removed and integrated into Chapter 1 when I'll have it sorted out. So, for the moment, you'll have to take it at face value, which, admittedly, feels a tad random.

4.1.1 Isometries and Killing Vectors

Consider a metric $g_{\mu\nu}$ and a given map of coordinates x_μ (which roughly translates into choosing a given reference frame). The metric will thus transform according to the standard tensor transformation rule

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}(x) \quad (2)$$

We'll say that a particular transformation $x \rightarrow x'$ is an *isometry* if

$$g'_{\rho\sigma}(y) = g_{\rho\sigma}(y) \quad (3)$$

If we consider an infinitesimal transformation of the likes

$$x'^\mu = x^\mu + \epsilon \xi^\mu \quad (4)$$

with $|\epsilon| \ll 1$, we can substitute directly into the transformation equation for the metric. Under the request of the transformation being an isometry, we'll obtain

$$\partial_\rho \xi^\mu g_{\mu\sigma} \partial_\sigma \xi^\nu g_{\rho\nu} + \xi^\mu \partial_\mu g_{\rho\sigma} = 0 \quad (5)$$

We can then introduce Christoffel's symbols to cast the equation in an equivalent form

$$\partial_\rho \xi_\sigma + \partial_\sigma \xi_\rho - 2\xi_\mu \Gamma_\rho^\mu{}_\sigma = 0 \quad (6)$$

The vector ξ^μ will be a *Killing vector* if it satisfies Killing's equation

$$\nabla_\rho \xi_\sigma + \nabla_\sigma \xi_\rho = 0 \quad (7)$$

⁵ It turned out that some white dwarves, low mass stars that had reached the end of their evolution cycle, would be older than the Universe. I hope you'll agree with me that it would be at least a bit odd.

If a given spacetime manifold has n spatial dimensions, then the space can have *at most* $n(n+1)/2$ Killing vectors. Such space is called *maximally symmetric*.

4.1.2 The Bare Minimum to Begin

Contemporary cosmological models are based on the idea the the Universe is pretty much the same everywhere, a stance sometimes known as the *Copernican Principle* or *Cosmological principle*. If we take it at face value, such a claim seems crazy, if at all logical: There's little resemblance, say, between the center of the Sun (which is kinda hot) and the desolate cold of the ISM. The Cosmological Principle starts to make a little more sense if we take it to apply only on the very largest scales, where local variations in density are averaged over and magically disappear.

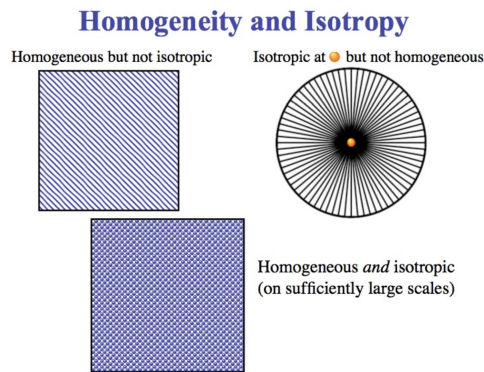


Figure 1: The Cosmological Principle automatically translates in two very specific requests: Homogeneity and Isotropy.

The Cosmological principle is related to two more mathematically precise properties that we can request our space-time manifold to have: *Isotropy* and *Homogeneity*, which are given a pictorial representation in Fig.1.

Homogeneity

Consider a family of hypersurfaces defined by some parameter t which allows us to foliate the manifold. Given any two points p and q in a manifold M , there is an isometry that takes p into q . See [1].

The assumption of Homogeneity essentially requires that the metric is the same throughout the manifold.

Isotropy

A manifold M is isotropic around a point p if, for any two vectors V and W in the tangent space of M to the point p $T_p M$, there is an isometry (symmetry) of M such that the *pushforward* of W under the isometry is parallel with V (not pushed forward). See [1].

Isotropy, instead, states that space looks exactly the same around a given point, no matter in what direction you're looking.

It turns out that a manifold that is simultaneously homogeneous and isotropic is also maximally symmetric! In fact the transformations associated to isotropy are the rotations

$$x'^{\mu} = R^{\mu}_{\nu} x^{\nu} \quad (8)$$

which generate $n(n-1)/2$ Killing vectors. On the other hand, the transformations associated to homogeneity are the translations

$$x'^{\mu} = x^{\mu} + \zeta^{\mu} \quad (9)$$

which generate n Killing vectors. The total number of Killing vector is, as expected, exactly $n(n+1)/2$. A spacetime which has a maximally symmetric spatial section must have a special metric that reflects such properties. Let's start from a general metric

$$ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} \quad (10)$$

Clearly, off-diagonal terms like g_{0i} must be vanishing $g_{0i} = 0$, or the request of an isotropic manifold wouldn't be met. This already leaves us with a metric with the following expression

$$ds^2 = -(u_{\mu} dx^{\mu})^2 + g_{ij}(t) dx^i dx^j \quad (11)$$

where the first term of the right-hand side represents the projection of the displacement along the tangent vector u^{μ} , which is, by virtue of its own definition, perpendicular to the spatial hypersurfaces.

Let's focus on the spatial metric $g_{ij}(t)$. Thanks to isotropy, the temporal dependence of the metric has to factor out, so that

$$g_{ij}(t) = a^2(t) \tilde{g}_{ij} \quad (12)$$

where we've introduced the *scale factor* $a(t)$.

Taking $u_{\mu} = (1, 0, 0, 0)$, our metric turns out to be

$$ds^2 = -dt^2 + a^2(t) \tilde{g}_{ij} dx^i dx^j \quad (13)$$

Let's now consider a n -dimensional geometry which we embed into a $n+1$ dimensional space.

3-Dimensional Space

So let's, for example, write down the metric for a 2-dimensional sphere embedded in a 3-dimensional space. The 3-dimensional Euclidean metric is

$$dl^2 = dx_1^2 + dx_2^2 + dx_3^2 \quad (14)$$

where we can define the radius of a given sphere a as

$$a^2 = x_1^2 + x_2^2 + x_3^2 \quad (15)$$

which we can implicitly differentiate to obtain an expression for, say, dx_3 which we'll plug into (14)

$$dx_3 = -\frac{x_1 dx_1 + x_2 dx_2}{\pm(a^2 - x_1^2 - x_2^2)^{1/2}} \quad (16)$$

so that the line element induced on the 2-sphere is

$$dl^2 = dx_1^2 + dx_2^2 + \frac{(x_1 dx_1 + x_2 dx_2)^2}{a^2 - x_1^2 - x_2^2} \quad (17)$$

We can define a vector $\vec{x} = (x_1, x_2)$, so that the line element can be expressed in a more compact fashion

$$dl^2 = d\vec{x}^2 + \frac{(\vec{x} \cdot d\vec{x})^2}{a^2 - \vec{x}^2} \quad (18)$$

Flat Space

The case of a flat space with vanishing curvature can be dealt with starting from the metric induced on the sphere, simply imposing $a \rightarrow \infty$. Not surprisingly, the resulting metric will be

$$dl^2 = d\vec{x}^2$$

Similarly, we can also calculate the induced metric on a 2-dimension hyperbola

$$a^2 = x_1^2 + x_2^2 - x_3^2 \quad (19)$$

The passages are essentially identical to those for the sphere, and the resulting metric is

$$dl^2 = d\vec{x}^2 - \frac{(\vec{x} \cdot d\vec{x})^2}{a^2 + \vec{x}^2} \quad (20)$$

For our future discussions it will be really useful to define a scaled-down x vector $\tilde{x} = \vec{x}/a$, so that the expression for the line element can be conveniently expressed as

$$dl^2 = a^2 \left(d\tilde{x}^2 + k \frac{(\tilde{x} \cdot d\tilde{x})^2}{1 - k\tilde{x}^2} \right) \quad (21)$$

where we've introduced the parameter $k = 0, \pm 1$, which defines the topological properties of the space we're considering.

Sometimes (read: almost always) it is useful to express the metric through the polar coordinates of the 2-dimensional sphere. With such purpose in mind, we're defining x_3 as the "height" of a certain circular section, r' the radius of such section, and θ the azimuthal angle with respect to axis parallel to x_1 . In formulas

$$\begin{aligned} x_3^2 &= a^2 - r'^2 \\ x_1 &= r' \cos \theta \\ x_2 &= r' \sin \theta \end{aligned}$$

With this new set of coordinates, the line element of the 2-dimensional sphere becomes

$$dl^2 = dr'^2 + r'^2 d\theta^2 + \frac{(r' dr')^2}{a^2 - r'^2} = a^2 \left(\frac{dr^2}{1 - r^2} + r^2 d\theta^2 \right) \quad (22)$$

where we've also introduced the adimensional radius $r = r'/a$, which lives in the interval $r \in [0, 1]$. We can repeat the calculations for the hyperbola

and the flat space in a similar fashion. The final result for the three scenarios we've considered can be once again expressed in a single expression thanks to the k parameter

$$dl^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 \right) \quad (23)$$

4-Dimensional Space

Perhaps it might have occurred to you that we do not, in fact, live in a 3-dimensional space, but rather in a 4-dimensional one with 3-dimensional spatial sections. The purpose of this section is essentially to replicate the calculations we've already went through for the 3-dimensional space but with one more additional dimension to address.

For this purpose, it comes somewhat in handy recalling (21), which allows us to immediately find the expression for the line element induced on a 3-dimensional surface embedded in a 4-dimensional space. All we have to do is replacing the metric of the 1-sphere S_1 ($d\theta^2$), with that of the 2-sphere S_2 , the solid angle $d\Omega_{(2)}^2$

$$dl^2 = a^2 \left(d\tilde{x}^2 + k \frac{(\tilde{x} \cdot d\tilde{x})^2}{1 - k\tilde{x}^2} \right) = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_{(2)}^2 \right) \quad (24)$$

For example, the metric induced on the sphere ($k = 1$) can be obtained defining the spherical coordinates (χ, θ, ϕ) and a radial distance a , so that

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = a^2 \quad (25)$$

and the line element is

$$dl^2 = dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2 \quad (26)$$

Performing the due substitutions you'd obtain

$$dl^2 = a^2 (d\chi^2 + \sin^2 \chi d\Omega_{(2)}^2) \quad (27)$$

The metric induced on the hyperboloid is obtained replacing $\sin \chi$ with $\sinh \chi$. Note that the new radial coordinate χ is adimensional, is in the range $[0, \pi]$, and is defined through

$$d\chi = \frac{dr}{\sqrt{1 - kr^2}} \quad (28)$$

In the hyperboloid case, the radial coordinate χ will now be in the range $[0, \infty)$, and the associated Cartesian coordinates are

$$\begin{aligned} x_1 &= a \sinh \chi \sin \theta \cos \phi \\ x_2 &= a \sinh \chi \sin \theta \sin \phi \\ x_3 &= a \sin \chi \cos \theta \\ x_4 &= a \cos \chi \end{aligned}$$

We can go through several boring substitutions to find the induced line element

$$dl^2 = a^2 (d\chi^2 + \sinh^2 \chi d\Omega_{(2)}^2) \quad (29)$$

and the compact expression for the three surfaces we're considering is

$$\boxed{dl^2 = a^2 \left(d\chi^2 + \frac{1}{k} \sin^2(\sqrt{k}\chi) d\Omega_{(2)}^2 \right)} \quad k = 0, \pm 1 \quad (30)$$

where we've made use of the fact that the line element can be conveniently expressed as

$$dl^2 = a^2 (d\chi^2 + f_k^2 d\Omega_{(2)}^2) \quad (31)$$

with f_k given by

$$f_k = \begin{cases} k^{-1/2} \sin(\sqrt{k}\chi) & k = 1 \\ \chi & k = 0 \\ (-k)^{-1/2} \sinh(\sqrt{-k}\chi) & k = -1 \end{cases} \quad (32)$$

which can be further compressed into

$$f_k = (k)^{-1/2} \sin(\sqrt{k}\chi) \quad (33)$$

which exploits the known limit for $k = 0$ for the sine and the definition of the hyperbolic sine for $k = -1$.

4.2 FRIEDMANN-ROBERTSON-WALKER METRICS

At this point we have all the elements needed to reintroduce time back into the picture

$$ds^2 = -dt^2 + a^2(t) \tilde{g}_{ij} dx^i dx^j = -dt^2 + dl^2 \quad (34)$$

If the Universe is spatially homogeneous and isotropic at all times, and if distances are allowed to expand or contract as a function of time only⁶, there's only one metric possible and is the FLRW metric:

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_{(2)}^2 \right) \quad (35)$$

Since $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, the elements of the metric tensor can be read directly from (35)

$$\begin{aligned} g_{00} &= -1 \\ g_{rr} &= \frac{a^2}{1 - kr^2} \\ g_{\theta\theta} &= r^2 a^2 \\ g_{\phi\phi} &= a^2 r^2 \sin^2 \theta \end{aligned}$$

while g_{0i} and off-diagonal terms of g_{ij} are vanishing.

From the coefficient of the metric, we can also compute Christoffel's symbols

$$\Gamma_{\mu\nu}^\alpha = \frac{1}{2} g^{\lambda\alpha} (\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \quad (36)$$

⁶ This is an inevitable consequence of the request of having a metric that is isotropic and homogeneous, for if all points didn't evolve at the same rate, it would be impossible for the Universe to be isotropic.

which have been reported in Appendix C.

Sometimes it is useful to perform a slight manipulation of the FLRW metric (35) and define a conformal time

$$a(\eta)d\eta = dt \quad (37)$$

so that the metric is, guess what, conformal.

Conformal Metric

Two metrics are conformal if there exists a conformal transformation such that

$$\tilde{g}_{\mu\nu} = \Omega(\eta)g_{\mu\nu}$$

and the metric $g_{\mu\nu}$ has no explicit dependence on the η parameter.

The scale factor a is therefore factorized and (35) becomes

$$ds^2 = a(\eta)^2 \left(-d\eta^2 + \frac{dr^2}{1 - kr^2} + r^2 d\Omega_{(2)}^2 \right) \quad (38)$$

Note that the quantity within parentheses is independent of the conformal time. On top of that, if $k = 0$ (flat space), the FLRW metric is conformal to a Minkowski metric, and the metric is said "conformally flat".

4.2.1 Geodesics

Consider a particle initially at rest in a FLRW metric (35). The geodesics equation for such a particle will be

$$\frac{d^2 x^i}{d\tau^2} = -\Gamma_{00}^i \frac{dx^0}{d\tau} \frac{dx^0}{d\tau} \quad (39)$$

since $dx^i/d\tau = 0$, where τ is the particle's proper time. However, since $\Gamma_{00}^i = 0$, the particle isn't accelerated and its radial coordinate remains constant!

Note that what remains constant is actually the particles *comoving coordinate*, while its *proper distance* calculated from some origin of the coordinate system changes because of the time-evolution of the scale factor a .

Once a system of axis is fixed so that a given point $P_1 = (r_1, 0, 0)$, we can define its proper distance as

$$d(r, t) = \int_0^{r_1} dl(r) \quad (40)$$

where $dl(r)$ is the line element induced on the manifold, in our case

$$dl^2 = a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_{(2)}^2 \right) \quad (41)$$

Ignoring the angular dependence, i.e. a point object for which $d\Omega = 0$, the proper distance is

$$d(r, t) = \int_0^{r_1} \frac{a(t) dr}{(1 - kr^2)^{1/2}} \quad (42)$$

which is solved by

$$d(r, t) = a(t) \begin{cases} \arcsin(r_1), & k = 1 \\ r_1, & k = 0 \\ \operatorname{arcsinh}(r_1), & k = -1 \end{cases} \quad (43)$$

Please note that unlike done by other textbooks, like [1], here the scale factor a has the dimension of a length rather than being adimensional. You can think of it as the result of the product between an adimensional scale factor (which is the one really changing) and the curvature radius of the Universe at some time.

Let us now consider a free particle with non-vanishing initial momentum. The general geodesics equation is

$$\frac{d^2 x^\sigma}{d\tau^2} = -\Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \quad (44)$$

It's useful to remember the definition of the 4-momentum $p^\mu = mu^\mu = m dx^\mu / d\tau$ (valid for timelike paths, at least), from which we know that

$$|\vec{p}|^2 = m^2 g_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau}$$

If we consider a system of coordinates such that at the instant considered the particles is arbitrarily close to the center of the coordinate system, the metric can thus be approximated as $\tilde{g}_{ij} = \delta_{ij} + o(x^2)$, where we're essentially neglecting the scalar product contribution to (21).

Then the Christoffel symbol $\Gamma_{jl}^i \propto \partial_l g_j^i \propto o(x)$ is negligible, since the space is always locally flat for each point we choose arbitrarily close to the origin.

The geodesics equation can be thus expanded

$$\begin{aligned} \frac{d^2 x^i}{d\tau^2} &= -\Gamma_{\mu\nu}^i \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -2\Gamma_{0j}^i \frac{dx^j}{d\tau} \frac{dx^0}{d\tau} \\ &= -2 \frac{\dot{a}(t)}{a(t)} \delta_j^i \frac{dx^j}{d\tau} \frac{dx^0}{d\tau} \end{aligned}$$

At this point we can multiply both terms by $d\tau/dt$

$$\frac{d^2 x^i}{d\tau^2} \cdot \frac{d\tau}{dt} = -2 \frac{\dot{a}(t)}{a(t)} \delta_j^i \frac{dx^j}{d\tau} \frac{dx^0}{d\tau} \cdot \frac{d\tau}{dt} \quad (45)$$

and exploit the identity

$$\frac{dx^0}{d\tau} \cdot \frac{d\tau}{dt} = 1 \quad (46)$$

resulting in a first-order differential equation for $dx^i/d\tau$

$$\frac{d}{dt} \left(\frac{dx^i}{d\tau} \right) = -2 \frac{\dot{a}(t)}{a(t)} \frac{dx^i}{d\tau} \quad (47)$$

This equation yields as a solution $dx^i/d\tau \propto a^{-2}$, which means that $|\vec{p}|^2 \propto m^2 a^2 \cdot a^{-4} \propto a^{-2}$. This means that a massive particle "slows down" with respect to the comoving coordinates as the Universe expands!

If we instead consider a null geodesic (massless particle), we would find out that

$$|\vec{p}| = p = \frac{\hbar}{\lambda} \implies \lambda \propto a(t) \quad (48)$$

which means that the wavelength of a photon changes depending on the time at which it was emitted, resulting in an effect commonly known as *cosmological redshift*.

4.2.2 A short detour on redshift

The purpose of this section is to show that we can obtain the same results that we've obtained in the last section using a more elegant approach that makes use of the symmetry properties of GR and the FLRW metric. Incidentally, this will also allow us to introduce a most useful quantity in Cosmology, which is the redshift z .

Consider the four-velocity of comoving observers $U^\mu = (1, 0, 0, 0)$, then the metric admits a Killing tensor of the form

$$K_{\mu\nu} = a^2(g_{\mu\nu} + U_\mu U_\nu) \quad (49)$$

that you can check⁷ that it satisfies $\nabla_{(\sigma} K_{\mu\nu)} = 0$ where ∇_σ is the covariant derivative. By definition of Killing vector (the generalization to a K. tensor is straightforward), the quantity

$$K^2 = K_{\mu\nu} V^\mu V^\nu = a^2 [V_\mu V^\mu + (U_\mu V^\mu)^2] \quad (50)$$

is conserved along the geodesics. Here $V^\mu = dx^\mu/d\lambda$. For massive particles, since $V_\mu V^\mu = -1$ (note the convention on the signature of the metric), we also have

$$(V^0)^2 = 1 + |\vec{V}|^2 \quad |\vec{V}|^2 = g_{ij} V^i V^j$$

Recalling the formula for the four-velocity of the observer, we have $U_\mu V^\mu = -V^0$, so (50) implies

$$|\vec{V}| = \frac{K}{a}$$

Notice how this means that the particle "slows down" with respect to the comoving coordinates as the Universe expands. If we instead consider null geodesics (that is what we're actually interested in)

$$V_\mu V^\mu = 0 \implies U_\mu V^\mu = \frac{K}{a}$$

but for null-like particles, like photons, we can write the four-velocity as $V^\mu = (\omega, \vec{k})$. So the frequency of the photon as measured by a comoving observer is $\omega = -U_\mu V^\mu$ (remember that $c = 1$). The frequency emitted will still be given by the same functional expression but calculated in the local inertial frame of the emitter, so $\omega_{em} = -K/a_{em}$. We then find out that

$$\frac{\omega_{obs}}{\omega_{em}} = \frac{a_{em}}{a_{obs}}$$

⁷ Perhaps it was trivial, but it took me a long while to figure out and convince myself of the proof, which you'll find in Appendix D.

To make things easier we then define a quantity called *redshift* defined as

$$z_{em} = \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} \quad (51)$$

so that, if the observation takes place today ($a_{obs} = a_0 = 1$), we can just write

$$a_{em} = \frac{1}{1 + z_{em}} \quad (52)$$

and

$$\frac{\nu_{obs}}{\nu_{em}} = \frac{1}{1 + z_{em}} \quad (53)$$

Let's give an example. If we observe, say, a galaxy with a redshift $z = 2$, we are observing it as it was when the Universe had a scale factor $a(t_e) = 1/3$.

The redshift we observe for a distant object thus depends only on the relative scale factors at the time of emission and the time of observation. It doesn't depend on how the transition between $a(t_e)$ and $a(t_0)$ was made: All that matters is the scale factors at the time of emission and the time of observation.

Calculating the redshift through spectroscopic observations, it was possible to correlate the recession velocity of objects in the Universe to their proper distance, obtaining the so called *Hubble's law*

$$v = cz = H_0 d_p \quad (54)$$

Recent measurements [4] managed to estimate the present-day Hubble constant H_0 with great precision

$$H_0 = 72.53 \pm 0.99 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (55)$$

from which we get an order-of-magnitude estimate of the age of the Universe

$$H_0^{-1} = 9.78h^{-1} \text{ Gyr} \quad (56)$$

using $H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1}$. We can also calculate the so-called Hubble radius

$$R_H = H_0^{-1}c = 3h^{-1} \text{ Gpc} \quad (57)$$

The variable h keeps track of the dispersion on the value of H_0 produced by the different measurement technique that was implemented.

Please note that Hubble's law is only valid on relatively large scales, where the peculiar velocity of objects become much smaller than the expansion velocity at a given distance.

On a concluding note, please note that although it's a useful meter of comparison to understand what we're talking about, cosmological redshift has *nothing to do* with a Doppler shift.

In fact, even if the Universe was in an age in which $\dot{a} = 0$ and so that $H_0 = 0$, you wouldn't be able to interpret the shift in frequency as the effect of a Doppler shift ($v = 0$ after all), yet you'd still observe the shift, since $a(t) \neq 0$.

4.2.3 Hubble-Le Maître's Law

Let's get back to the definition of redshift starting from the scale factor a which we can Taylor expand for emission times t "close" to the observation time

$$a(t) = a(t_0) + \left(\frac{\partial a}{\partial t} \right)_{t_0} (t - t_0) + o((t - t_0)^2) \quad (58)$$

so that

$$a(t) = a(t_0) [1 + H_0(t - t_0) + o((t - t_0)^2)] \quad (59)$$

The expansion won't be ill-defined only if $H_0^{-1} \gg (t - t_0)$. We can now plug (59) into the definition of redshift (52)

$$z \approx H_0(t_0 - t) \implies z = \frac{H_0 d}{c} \quad (60)$$

where $d = c(t_0 - t)$ is the distance spanned by photons in the same time interval. Equation (60) is the correct way to express Hubble-Le Maître's law for not too large distances. For larger distances we would have to take into account the modification brought about by the second order term in the expansion of a , forcing us to introduce an additional parameter, the *acceleration parameter* q .

4.3 THE ROAD TO FRIEDMANN'S EQUATIONS

Conserved current for a non-collisional fluid

For the time being, let's consider a Universe completely filled with a fluid of cosmological relevance with negligible collision rate. The number of particles that make the fluid is conserved, and since the fluid must obey the Cosmological Principle, collective motion of the fluid can't have a privileged direction, hence $\langle v^i \rangle = 0$. We can thus define a 4-current vector J^μ that we will shortly prove to be conserved

Please note that we're walking, as usual, with units in which $c = 1$.

$$J^\mu = n u^\mu = n \frac{dx^\mu}{d\tau} \quad u^\mu = (u^0, \gamma \vec{v}) \quad (61)$$

where n is the number density of particles. To see how the expansion of the Universe affects the number density n we have to first compute

$$\nabla_\mu J^\mu = \partial_\mu J^\mu + \Gamma_\mu{}^\mu{}_\nu J^\nu \quad (62)$$

But since the metric is the FLRW, the covariant divergence of J^μ is vanishing. In fact, since J^i is zero on average, the covariant divergence is reduced to simplu

$$\partial_0 J^0 + \Gamma_i{}^i{}_0 J^0 = 0 \quad \Gamma_i{}^i{}_0 = \frac{\dot{a}}{a} \delta^i{}_i = 3 \frac{\dot{a}}{a} \quad (63)$$

which means

$$\frac{dn}{dt} = -3 \frac{\dot{a}}{a} \implies n(t) \propto a^{-3} \quad (64)$$

4.3.1 Energy-Momentum Tensor and Its Conservation

For the sake of simplicity, let's consider first the case of a flat-space. The energy-momentum tensor can be introduced from the minimal-action principle, imposing that the variation of a properly defined action functional is vanishing *on shell*⁸

$$\delta S = \delta \left[\int dt dV \mathcal{L} \left(q^{(a)}, \frac{\partial q^{(a)}}{\partial x^\mu} \right) \right] = 0 \quad (65)$$

where $\mathcal{L}$ is a lagrangian density. Computing the variation we obtain Euler-Lagrange equations

$$\partial^\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial^\mu q)} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0 \quad (66)$$

Exploiting Noether's theorem and assuming translational invariance for the lagrangian density we could show that there exists a conserved quantity representing the 4-momentum flux crossing the fixed time hypersurfaces, whose normal vector is tangent to the flux itself

$$\partial_\nu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\nu q)} \partial_\mu q - \delta_\mu^\nu \mathcal{L} \right] = \partial_\nu T_\mu^\nu = 0 \quad (67)$$

The 4-momentum is thus simply defined as the flux of the energy-momentum tensor through the hypersurfaces

$$p^\mu = \int T^{\mu\nu} dS_\nu \quad dS_\nu = dV \hat{n}_\nu \quad (68)$$

where $\hat{n}_\nu = (1, 0, 0, 0)$ is the unit vector orthogonal to the fixed-time hypersurfaces. For reasons that I'm sure won't surprise you, T^{00} is the energy density, while T^{i0} is the momentum density. Moreover, you can easily check that

$$\int_V \partial_\alpha T^{i\alpha} dV = 0 \implies \partial_t \int_V T^{i0} dV + \int_V \partial_j T^{ij} dV = 0 \quad (69)$$

Applying the divergence theorem to the second integral yields

$$\partial_t \int_V T^{i0} dV + \int_{\partial V} T^{ij} dS_j = 0 \quad (70)$$

which means that forces must be evenly balanced by the pressure arising in the fluid.

When a fluid is isotropic and homogeneous in a comoving reference frame, the fluid is said to be *perfect*. Through long, and boring, calculations (which essentially consist in transforming T^{00} and T^{ij} out of the comoving frame), you can show that a perfectly valid expression for the energy-momentum tensor of a perfect fluid is

$$T^{\mu\nu} = (p + \rho) u^\mu u^\nu + p g^{\mu\nu} \quad (71)$$

which we can also present in the form of a (1,1) tensor

$$T^\mu_\nu = g_{\alpha\nu} T^{\mu\alpha} = (p + \rho) u^\mu u_\nu + p (g^{\mu\alpha} g_{\nu\alpha}) \quad (72)$$

⁸ i.e. when calculated on the equations of motion.

Note that the components of u^μ are fixed by requiring $u_\mu u^\mu = -1$

from which we can immediately compute the trace

$$T^\mu_\mu = -(p + \rho) + 4p = 3p - \rho \quad (73)$$

What we'd like to show now is that the energy-momentum tensor (71) we've defined is conserved $\nabla_\nu T^{\mu\nu} = 0$ also if the metric is no longer that of the flat space but rather the FLRW (35). To do so, we have to prove that

$$\nabla_\nu T^{\mu\nu} = \nabla_\nu [(p + \rho)u^{\mu\nu}] + \nabla_\nu [pg^{\mu\nu}] = 0 \quad (74)$$

After exploiting metric compatibility, and expanding the covariant derivative, we can rewrite the previous expression as

$$\partial_\nu (p + \rho)u^\mu u^\nu + \partial_\nu pg^{\mu\nu} + (p + \rho) \left[\Gamma^\mu_{\nu\lambda} u^\lambda u^\nu + \Gamma^\nu_{\nu\lambda} u^\mu u^\lambda \right] \quad (75)$$

At this point we can notice that, since the fluid is comoving, $u^\mu = (1, 0, 0, 0)$, and thus

$$\Gamma^\mu_{\nu\lambda} u^\lambda u^\nu = \Gamma^\mu_{00} u^0 u^0 \quad \Gamma^\nu_{\nu\lambda} u^\mu u^\lambda = \Gamma^\nu_{\nu 0} u^\mu u^0 \quad (76)$$

But $\Gamma^\mu_{00} = 0$ for all values of μ given the FLRW metric, while the second term can be simplified as

$$\Gamma^\nu_{\nu 0} u^\mu u^0 = \frac{1}{2} g^{\nu i} \partial_0 g_{\nu i} u^\mu \quad (77)$$

Let's focus on the purely spatial component $\mu = j$, which ultimately simplifies to just $\partial_i pg^{ij}$, which is vanishing since the pressure has to be homogeneous because of the Cosmological Principle. The only non-trivial component is the temporal one $\mu = 0$, which yields

$$\partial_0(p + \rho) - \partial_0 p + \frac{p + \rho}{2} g^{ij} \partial_0 g_{ij} \quad (78)$$

The last derivative can be hand-wavily calculated expanding $g_{ij} = a^2 \tilde{g}_{ij}$, where $\tilde{g}_{ij}$ is independent of time. This way, the temporal component ultimately yields

$$\boxed{\partial_0 \rho + 3(p + \rho)H(t) = 0} \quad (79)$$

It might be worth pointing out that ρ is here an energy density rather than a matter density.

4.3.2 Ricci Tensor in a FLRW Metric

Honestly, I'm not in the mood to calculate all non-trivial components of the Riemann tensor first

$$R^\mu_{\nu\alpha\beta} = \partial_\alpha \Gamma^\mu_{\nu\beta} - \partial_\beta \Gamma^\mu_{\nu\alpha} + \Gamma^\mu_{\sigma\alpha} \Gamma^\sigma_{\nu\beta} - \Gamma^\mu_{\sigma\beta} \Gamma^\sigma_{\nu\alpha} \quad (80)$$

to then calculate Ricci's $R_{\nu\beta} = R^\alpha_{\nu\alpha\beta}$. You can find all non-trivial components of the Ricci tensor in Appendix C. Without doing all the troublesome calculations, I'll just say that in principle the calculation is made a little easier if we consider small deviations from flat-space for the time independent metric $\tilde{g}_{ij} = g_{ij}/a^2$, which helps significantly to decrease the effort.

It is worth mentioning, however, that we might as well expect the components R_{0i} to be vanishing because of the request of homogeneity and isotropy. An elegant way of proving it without losing your wits in the process might be considering the "inverted" version of Einstein's equation

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) \quad (81)$$

where T is the trace of the energy-momentum tensor. From here is trivial to prove our previous claim if we assume $T_{\mu\nu}$ to be the energy momentum tensor of a perfect fluid as seen by a comoving observer.

For future reference, I'm going to write the non-trivial components of the Ricci tensor down below

$$R_{00} = -\frac{3\ddot{a}}{a} \quad R_{ij} = \frac{1}{a^2} (2k + a\ddot{a} + 2\dot{a}^2) g_{ij} \quad (82)$$

4.3.3 Friedmann's Equation

Consider the non-canonical form of Einstein's equation (81). The trace of the energy momentum tensor is (71), and its trace is $T = 3p - \rho$, so it's immediate to solve Einstein's equation.

The temporal component yields

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)} \quad (83)$$

Since we have three independent variables and only two viable equations⁹, we're in the need of introducing an equation of state (EoS) that relates the pressure to the energy density. Cosmology usually deals with dilute gases, for which the equation of state is simple.

For substances of cosmological importance

$$p = w\rho \quad (84)$$

where w is a dimensionless number. The behaviors of our favorite sources are summarized in the following table

	w_i
matter	0
radiation	$\frac{1}{3}$
curvature	$-\frac{1}{3}$
vacuum	-1

With our newly gained EoS in mind, we can rewrite Friedmann equation as

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho(1 + w) \quad (85)$$

⁹ The equations are actually 3: Two Friedmann equations and the equation coming from the conservation of the temporal component of the energy-momentum tensor. The three of them are not independent, since they're related through Bianchi's identity.

which tells us that the Universe is accelerating if and only if

$$\frac{\ddot{a}}{a} > 0 \iff w < -\frac{1}{3} \quad (86)$$

The spatial component of (81), on the other hand, yields

$$\frac{2k}{a^2} + \frac{\ddot{a}}{a} + 2 \left(\frac{\dot{a}}{a} \right)^2 = -4\pi G(p - \rho) \quad (87)$$

where the common factor $\tilde{g}_{ij}$ was suppressed. It is customary however to subtract the 1st Friedmann equation (83) from the equation we've just derived, obtaining the 2nd Friedmann equation

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} \quad (88)$$

Suppose that the Universe contains N different components, with the i -th component having an energy density ρ_i and an EoS parameter w_i . As long as there is no interaction between the different components, the fluid equation (79) must hold for each component separately. So

$$\frac{\dot{\rho}_i}{\rho_i} = -3(1 + w_i) \frac{\dot{a}(t)}{a(t)} \quad (89)$$

which can be immediately integrated. Behold:

$$\rho_i(a) = \rho_{i,0} \left(\frac{a(t)}{a(t_0)} \right)^{-3(1+w_i)} \quad (90)$$

Equation (90) implies that for (non-relativistic) matter $\rho \propto a^{-3}$, while for radiation (or relativistic matter) $\rho \propto a^{-4}$. This fact has a simple explanation: The energy density of non-relativistic matter decreases as a increases because of the expanding volume of the Universe, while relativistic matter also gets redshifted. This has also another striking consequence: There had been times when different components were equivalent!

$$\begin{aligned} \frac{\rho_\Lambda(a)}{\rho_m(a)} = 1 &= \frac{\rho_{\Lambda,0}}{\rho_{m,0}} a^3 \implies a_{\Lambda,m} = \left(\frac{\rho_{m,0}}{\rho_{\Lambda,0}} \right)^{1/3} \approx 0.766 \implies z_{\Lambda,m} = 0.31 \\ \frac{\rho_m(a)}{\rho_r(a)} = 1 &= \frac{\rho_{m,0}}{\rho_{r,0}} a \implies a_{r,m} = \frac{\rho_{r,0}}{\rho_{m,0}} \approx 2.9 \cdot 10^{-4} \implies z_{r,m} = 3400 \end{aligned}$$

If the Universe contains different components with different values of w , in the limit $a \rightarrow 0$, the component with the largest value of w is dominant.

Now that we have a relation between ρ and a we can set to solve the Friedmann equation (88) in special case of a spatially flat ($k = 0$) and single-component Universe. Plugging in (90), we find

$$\dot{a}^2 = \frac{8\pi G \rho_0}{c^2} \frac{a^{-(1+3w)}}{a_0^{-(1+3w)}} \quad (91)$$

which can be solved explicitly

$$a(t) = a_0 \left(\frac{t}{t_0} \right)^{\frac{2}{3+3w}} \quad t_0 = \frac{2}{3H_0(1+w)} \quad w \neq -1 \quad (92)$$

For $w = -1$, the equation is solved by

$$a(t) = a_0 e^{\pm H_0(t-t_0)} \quad (93)$$

A detour on conformal time

Let's now write Friedmann's equations using the conformal time $d\eta = dt/a$. Through a basic application of the chain rule

$$\frac{da}{dt} = \frac{da}{d\eta} \frac{d\eta}{dt} = a' \frac{1}{a} \quad (94)$$

where with the superscript we're indicating derivatives with respect to the conformal time. In short

$$a' = \dot{a}a \quad a'' = a(a\ddot{a} + \dot{a}a') \quad \mathcal{H} = \frac{a'}{a} = \dot{a} \quad \mathcal{H}' = a\ddot{a} \quad (95)$$

With these new variables, Friedmann's equation become

$$\mathcal{H}^2 = \frac{8\pi G}{3} a^2 \rho - k \quad (96)$$

$$\mathcal{H}' = -\frac{4\pi G}{3} a^2 (\rho + 3p) \quad (97)$$

while the conformal fluid equation is

$$\rho' + 3\mathcal{H}(p + \rho) = 0 \quad (98)$$

Part I

APPENDIX

APPENDIX B: BLACKBODY FUNCTION

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu) d\nu = \frac{4\pi h g \nu^3}{c^3} d\nu$$

from which deducing the expression from internal energy is straightforward. Remembering $g = 2$ is the quantum degeneracy of photons, a simple multiplication of the previous expressions yields

$$U_\nu d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{\exp(h\nu/kT) - 1} d\nu \quad (99)$$

The energy of a photon is $E_\gamma = h\nu$, meaning the number of photons in the frequency range $\nu \rightarrow \nu + d\nu$ is

$$n_\nu d\nu = \frac{8\pi}{c^3} \frac{\nu^2}{\exp(h\nu/kT) - 1} d\nu \quad (100)$$

which can be integrated over all frequencies, thus obtaining the number density of photons in blackbody radiation

$$n_\gamma = \beta T^3 \quad \beta = \frac{2.4041k^3}{\pi^2 \hbar^3 c^3} \quad (101)$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/kT) - 1} \quad (102)$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but it's possible to generalize this formulation for less "restrictive" environments.

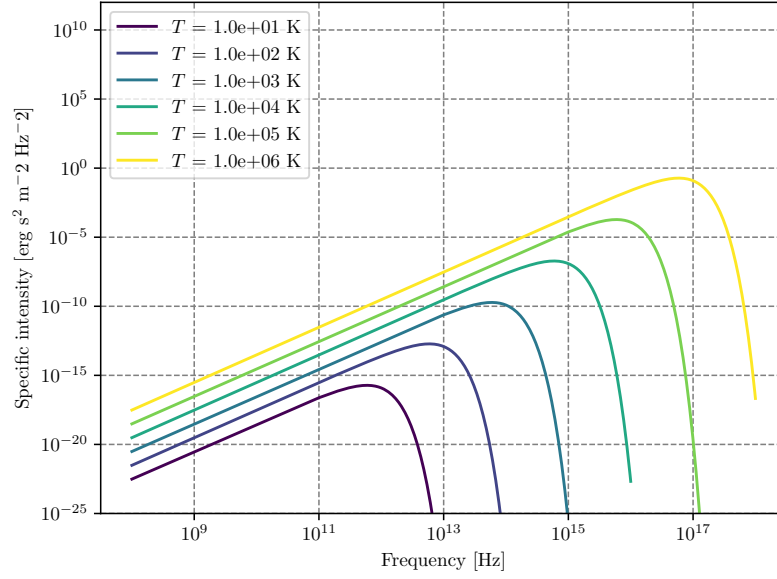


Figure 2: Blackbody frequency spectrum.

An incredible number of important results descends from (102), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T) d\nu = \frac{2h}{c^2} \frac{\pi^4}{15} \left(\frac{kT}{h} \right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example¹

$$F(T) = \pi \int_0^{+\infty} B_\nu(T) d\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2 h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2 h^3} T^4 \tag{103}$$

is what is usually known as the *Stefan-Boltzmann law*.

¹ The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.102: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2} kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^W = \frac{2h\nu^3}{c^2} \exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars. The definitions are those of [5].

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k} I_\nu \quad h\nu/kT \ll 1 \quad (104)$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν .

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν , from an unresolved source we cannot find I_ν , unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c , is obtained. Often the “fitting” procedure is nothing more than estimating the peak of the spectrum and applying Wien’s displacement law to find a temperature.

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos \theta \, d\nu \, d\Omega := \sigma T_{eff}^4 \quad (105)$$

APPENDIX C: CHRISTOFFEL SYMBOLS

For a symmetric, metric-compatible covariant derivative ∇_μ , the Christoffel symbols are given by

$$\Gamma_{\mu\nu}^\alpha = \frac{1}{2}g^{\lambda\alpha}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \quad (106)$$

For the FLWR metric (35), the Christoffel symbols are given by

$$\begin{aligned} \Gamma_{11}^0 &= \frac{a\dot{a}}{1-kr^2} & \Gamma_{11}^1 &= \frac{kr}{1-kr^2} \\ \Gamma_{22}^0 &= a\dot{a}r^2 & \Gamma_{33}^0 &= a\dot{a}r^2 \sin^2 \theta \\ \Gamma_{0j}^i &= \frac{\dot{a}}{a}\delta_j^i = H\delta_j^i & \Gamma_{01}^1 &= \frac{\dot{a}}{a} = H \\ \Gamma_{22}^1 &= -r(1-kr^2) & \Gamma_{33}^1 &= -r(1-kr^2) \sin^2 \theta \\ \Gamma_{12}^2 &= \Gamma_{13}^3 = \frac{1}{r} & \Gamma_{33}^2 &= -\sin \theta \cos \theta \\ \Gamma_{23}^3 &= \cot \theta & \Gamma_{00}^i &= 0 \\ \Gamma_{00}^0 &= 0 & \Gamma_{i0}^0 &= 0 \end{aligned}$$

or related to this by symmetry. The non-zero components of the Ricci tensor are

$$\begin{aligned} R_{00} &= -3\frac{\ddot{a}}{a} \\ R_{11} &= \frac{a\ddot{a} + 2\dot{a}^2 + 2k}{1-kr^2} \\ R_{22} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k) \\ R_{33} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k) \sin^2 \theta \end{aligned}$$

and the Ricci scalar is then

$$R = 6 \left[\frac{\ddot{a}}{a} \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right]$$

APPENDIX D: WHERE THE D STANDS FOR DEMONSTRATION

The FLRW Killing Tensor

In Chapter 4 we've defined at a certain point a Killing tensor (49) but I never showed that it actually satisfies Killing's equation. Our goal is to show that the tensor

$$K_{\mu\nu} = a^2(g_{\mu\nu} + u_\mu u_\nu) \quad (107)$$

satisfies $\nabla_{(\sigma} K_{\mu\nu)} = 0$.

Let's start from observing that $K_{\mu\nu}$ is a spatial projector

$$u^\nu K_{\mu\nu} = a^2(u^\nu g_{\mu\nu} + u_\mu u_\nu u^\nu) = 0 \quad (108)$$

since $u_\nu u^\nu = -1$. This means that we can define $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ so that $K_{\mu\nu} = a^2 h_{\mu\nu}$. Note that $h_{\mu\nu}$ coincides with the spatial sections of the FLRW metric.

Before calculating the covariant derivative of the alleged Killing tensor, let's first check

$$\begin{aligned} \nabla_\mu u_\nu &= \partial_\mu u_\nu - \Gamma_{\mu\nu}^\lambda u_\lambda \\ &= +\Gamma_{\mu\nu}^0 = \frac{1}{2} g^{00} \partial_0 g_{\mu\nu} \\ &= -\frac{1}{2} \partial_0 (h_{\mu\nu} - u_\mu u_\nu) \\ &= -\frac{1}{2} \partial_0 h_{\mu\nu} = -\frac{1}{2} \partial_0 (a^2 \tilde{h}_{\mu\nu}) = -a\dot{a} \tilde{h}_{\mu\nu} = -H h_{\mu\nu} \end{aligned}$$

We can now expand the covariant derivative of $K_{\mu\nu}$

$$\begin{aligned} \nabla_\lambda K_{\mu\nu} &= h_{\mu\nu} \nabla_\lambda a^2 + a^2 \nabla_\lambda h_{\mu\nu} \\ &= 2a\dot{a} u_\lambda h_{\mu\nu} + a^2 \nabla_\lambda h_{\mu\nu} \\ &= 2a\dot{a} u_\lambda h_{\mu\nu} + a^2 (u_\mu \nabla_\lambda u_\nu + u_\nu \nabla_\lambda u_\mu) \end{aligned}$$

where we can now substitute $\nabla_\mu u_\nu = -H h_{\mu\nu}$ to obtain

$$\begin{aligned} \nabla_\lambda K_{\mu\nu} &= 2a\dot{a} u_\lambda h_{\mu\nu} - a\dot{a} (u_\mu h_{\nu\lambda} + u_\nu h_{\mu\lambda}) \\ &= a\dot{a} [(u_\lambda h_{\mu\nu} - u_\mu h_{\lambda\nu}) + (u_\lambda h_{\mu\nu} - u_\nu h_{\lambda\mu})] \\ &= 2a\dot{a} [u_{[\lambda} h_{\mu]\nu} + u_{[\lambda} h_{\nu]\mu}] \end{aligned}$$

At this point, we can easily see that the symmetrization $\nabla_{(\sigma} K_{\mu\nu)}$ of the previous expression yields zero, thus proving that $K_{\mu\nu}$ is indeed a Killing tensor.

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Bachelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

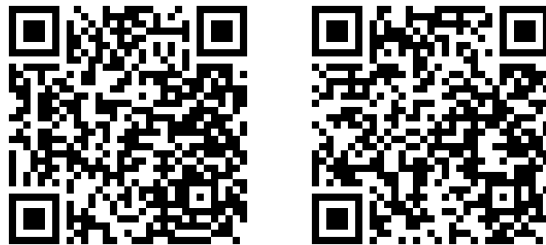


Figure 3: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my colleagues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

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